

Data Landscape in Current Generation:

1. The Data Maturity Model



Maslow's hierarchy of needs

In terms of data out hierarchy model becomes:

1. Data collection
2. Data wrangling
3. Data integration
4. BI and Analytics
5. Artificial Intelligence

2. ETL and ELT

ETL or Extract Transform and Load is process where we first extract data , then transform the data and finally load it into our data targets.

ELT or Extract Load Transform is process where we extract from source, Load it to preserve history and for other applications and then do necessary transformations.

3. Data warehousing:

Data warehousing is collection of organized and streamlined data storage solution and Generally ELT is preferred

Some of data warehouses are Snowflake, Redshift , Big Query etc.

4. Data Lake:

Data lakes is dump of all data that contains all raw data and generally ETL is preferred.

Some data lakes are Delta Lake, Data Lake, Databricks etc.

Slowly Changing Dimensions:

SCD is an alternative to CDC where history data is retained

1. **Type-0:** Only Source table is changed no need to change the data warehouse.
2. **Type-1 :** Updating the table when a dimension changes, overwriting the original table.
3. **Type-2 :** Keeping full history -Adding additional rows for each dimension change.
4. **Type-3 :** Keeping limited data history- Adding Separate columns for original and current values are retained.